Jark-Wah Andrew Wong

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EDUCATION

The Pennsylvania State University — **State College, PA**
Doctor of Philosophy in Chemical Engineering — Aug. 2019 - Present
Doctoral Minor in Computational Materials — **Anticipated Graduation – June 2024**
Advisor: Dr. Michael J. Janik — GPA: 3.87/4.0

Texas A&M University — **College Station, TX**
Bachelor of Science in Chemical Engineering, *magna cum laude* — Aug. 2015 – May 2019
Certification: Engineering Therapeutics Manufacturing — GPA: 3.7/4.0
Dean's List (2016-2019)

RESEARCH EXPERIENCE

Computational Chemistry and Materials Science Summer Institute Internship
Lawrence Livermore National Laboratory, Dr. Sneha Akhade — May 2023 – Present
Collaboration: Dr. Christopher Hahn of Lawrence Livermore National Laboratory

- Guiding computational efforts between multi-disciplinary group for the design of biomass upgrading electrocatalysts
- Utilizing DFT methods to study ligand and strain effects during the electro-oxidation of HMF and alcohols in biomass feedstocks across Ni oxide surfaces supported on Au or Ag underlayers

Understanding the dependence between the activity of CO_2 electro-reduction, identity of alkali metal cations, and surface facets of Au within the electrode-electrolyte interface
The Pennsylvania State University, Dr. Michael J. Janik — Mar. 2023 – Present
Collaboration: Dr. Anne Co of Ohio State University

- Leading computational efforts using DFT/FF-MD methods to better understand the impacts of the electrolyte's ions and metal facet dependence during CO_2 electrochemical reduction on Au surfaces
- Studying changes in EDL interfacial properties due to different electrolyte compositions and surface facets using large scale FF-MD simulations

Investigating anion adsorption of phosphonic and sulfonic acid ionomer binders in electrochemical hydrogen pumps
The Pennsylvania State University, Dr. Michael J. Janik — Feb. 2023 – Nov. 2023
Collaboration: Dr. Chris Arges of Penn State University

- Studying correlations between the adsorption affinity of phosphonic and sulfonic acidic anions and experimental charge resistance values during HER/HOR kinetics in electrochemical hydrogen pumps using DFT methods

Electro-oxidative valorization of biomass: design strategies for selective and stable catalysis
The Pennsylvania State University, Dr. Michael J. Janik — Nov. 2022 – Present
Collaboration: Dr. Adam Holewinski of University of Colorado at Boulder

- Determining fundamental activity and selectivity descriptors for the design of oxidation bimetallic electrocatalysts for biomass upgrading of multi-carbon organic molecules using DFT methods
- Deciphering vibrational frequencies of surface-bound furfural intermediates from experimental ATR-SEIRAS spectra using DFT methods

DFT and Classical MD (FF-MD) methods to study electrochemical double-layers and electrode-electrolyte effects
The Pennsylvania State University, Dr. Michael J. Janik Jan. 2020 – Present
Collaboration: Dr. Scott T. Milner of The Pennsylvania State University

- Developing a combined DFT/FF-MD approach to model electrochemical double layer (EDL) structures and its effects on electrocatalytic barriers, adsorption, and electrocatalytic reaction energetics
- Probing specific alkali metal cation adsorption across late transition metal surfaces and sensitivities of DFT prediction of thermodynamic equilibria due to approximation of the EDL using DFT and FF-MD methods

Investigating the electrocatalytic reduction mechanism of nitroaromatics using DFT methods
The Pennsylvania State University, Dr. Michael J. Janik Jan. 2020 – May 2022
Collaboration: U.S. Army, Catalyze H_2O, and Dr. Lauren Greenlee of The Pennsylvania State University

- Evaluated and screened potential TNT and nitrobenzene reduction electrocatalysts across late transition metal and metal oxide surfaces for the environmental treatment of wastewater using DFT methods

Developing nanoplasmonic sensors coupled with Surface Enhanced Raman Spectroscopy for detection applications
Texas A&M University, Dr. Hung-Jen Wu Dec. 2017 – May 2019

- Experimentally synthesized Ag-based catalysts through vapor deposition and Tollens' test
- Characterized Ag deposited catalysts using Raman spectroscopy for drug detection applications

RESEARCH PUBLICATIONS

Peer-review papers

Z. Cui, **A. J. Wong,** M. J. Janik, and A. Co. *"The negative reaction order of CO was determined during CO_2* electroreduction on Au". Manuscript in preparation.

A. J. Wong, B. H. Tran, S. T. Milner, and M. J. Janik. "*Incorporating electrode-electrolyte interfacial effects during specific cation adsorption on late transition metals using Density Functional Theory methods*". Manuscript in preparation.

J. L. Miller, **A. J. Wong**, and M. J. Janik. "*Electrocatalytic properties of an iron organometallic complex and its role in the reduction of nitroaromatic compounds*". Manuscript in preparation.

A. J. Wong*, B. H. Tran*, N. Agrawal, B. R. Goldsmith, and M. J. Janik. *"Sensitivity Analysis of Electrochemical Double Layer Approximations on Electrokinetic Predictions: Case Study for CO Reduction on Copper"*. (Review in Journal of Physical Chemistry C).

Z. Cui*, **A. J. Wong*,** M. J. Janik, and A. Co. *"Cation effects on CO_2 reduction by single-crystal and polycrystalline gold electrodes under well-defined mass transport conditions*". (Review in Journal of the American Chemical Society).

N. Agrawal, **A. J. Wong**, S. Maheshwari, and M. J. Janik. *"An efficient approach to compartmentalize double layer effects on kinetics of interfacial proton-electron transfer reactions"*. **Journal of Catalysis**, February 2024

K. Arunagiri, **A. J. Wong**, L. Briceno-Mena, J. A. Romagnoli, M. J. Janik, and C. G. Arges. *"Deconvoluting charge-transfer, mass transfer, ohmic resistances in phosphonic acid-sulfonic acid ionomer binders used in electrochemical hydrogen pumps"*. **Energy and Environmental Science**, 2023, Advance Article, October 2023

A. J. Wong, J. L. Miller, B. Perdue, and M. J. Janik. "*Investigating the electrocatalytic reduction of 2,4,6-tri-nitro-toluene (TNT) using Density Functional Theory methods*". **Green Chem.** Advanced Article, May 2023

A. J. Wong, J. L. Miller, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late-transition-metal surfaces from Density Functional Theory*". **Chem Catalysis**. Vol 2, Issue 6, P1362-1369, June 2022

* = equal contribution as first co-authors

Book Chapters

N. Agrawal, K. Yeh, S. Maheshwari, B. Tran. **A. J. Wong**, and M. J. Janik. *"Density Functional Theory Methods for Electrocatalysis."* Book chapter in preparation.

Oral Presentations

AIChE Fall 2024, San Diego, Nov. 2024. B. Tran, A. J. Wong, M. J. Janik, and B. Goldsmith. *"Theoretical Examination of Potential-Dependent CO Adsorption and Reaction on Copper Electrodes."*

ACS Spring 2024, New Orlean, Mar. 2024. A. J. Wong, Z. Cui, A. C. Co, and M. J. Janik. *"Electrode-electrolyte effects on electrocatalytic kinetics using a compartmentalized DFT and double-layer theory approach."* **(Invited Talk)**

AIChE Annual Meeting, Orlando, Nov. 2023. **A. J. Wong**, B. Tran, N. Agrawal, D. R. Zhu, S. T. Milner, and M. J. Janik. *"Multi-Scale Modeling of Electrocatalytic Processes within the Electrochemical Double Layer."*

NAM 28th Annual Meeting, Rhode Island, June 2023. **A. J. Wong,** B. Tran, S. T. Milner, and M. J. Janik. *"Modeling Specific Alkali Cation Adsorption and Electrochemical Double-Layers with DFT and Classical MD."*

AIChE Annual Meeting, Phoenix, Nov. 2022. **A. J. Wong**, B. Tran N. Agrawal, S. T. Milner, and M. J. Janik. "*Incorporating Electrode-Electrolyte Interfacial Effects on the Specific Adsorption of Ions on Late Transition Metal Surfaces Using a Combined DFT/FF-MD Approach*."

AIChE Annual Meeting, Phoenix, Nov. 2022. B. Tran, **A. J. Wong**, N. Agrawal, S. T. Milner, and M. J. Janik. "*A Combined Classical MD and DFT Approach for Modeling Ionic Adsorption on Metal Electrodes with Explicit Description of the Electrical Double Layer*."

PSU ChE Research Symposium, University Park, Sept. 2022. **A. J. Wong,** B. Tran, S. T. Milner, and M. J. Janik. "*Incorporating Electrode-Electrolyte Interfacial Effects on the Specific Adsorption of Ions on Late Transition Metal Surfaces Using a Combined DFT/FF-MD Approach*."

ACS Spring 2022 Meeting, San Diego, March 2022. **A. J. Wong.** J. L. Miller, B. Perdue, and M. J. Janik. *"Elementary mechanisms of the electrocatalytic reduction of nitroaromatics on late transition metal surfaces using Density Functional Theory methods."*

AIChE Annual Meeting, Boston, Nov. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. *"Mechanistic Studies on the Electrocatalytic Reduction of Nitroaromatic Compounds."*

Poster Presentations

AIChE Annual Meeting, Orlando, Nov. 2023. **A. J. Wong**, B. Tran, M. J. Janik. "*Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach*." Poster Presentation Accepted.

2023 Catalysis Club of Philadelphia Annual Symposium-Student Poster Joint Conference. Philadelphia, May. 2023. **A. J. Wong**, B. H. Tran, D. R. Zhu, S. T. Milner, and M. J. Janik. **"***Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach***."**

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, May. 2023. **A. J. Wong**, B. H. Tran, D. R. Zhu, S. T. Milner, and M. J. Janik. **"***Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach***."**

AIChE Annual Meeting, Phoenix, Nov. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Investigating the Electrocatalytic Reduction of 2,4,6-Tri-Nitro-Toluene (TNT) across Late Transition Metal Surfaces Using Density Functional Theory Methods.*"

PSU ChE Research Symposium, University Park, Sept. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, Aug. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

AIChE Annual Meeting, Boston, Nov. 2021. J. L. Miller, B. Perdue, **A. J. Wong**, and M. J. Janik. "Electrocatalytic Reduction of Nitrobenzene by Iron-Salen Ligand Complexes"

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, Oct. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

PSU ChE Research Symposium, University Park, Oct. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

TEACHING & WORK EXPERIENCE

Graduate Teaching Assistant | The Pennsylvania State University Aug. 2022 – Present
Course: Graduate Chemical Reaction Engineering

- Teaching advanced graduate chemical reaction engineering principles to first year graduate students
- Provide mentorship and guidance for incoming first-year chemical engineering graduate students

Graduate Teaching Intern and Assistant | The Pennsylvania State University Jan. 2022 – May 2022
Course: Undergraduate Chemical Reaction Engineering

- Led and guided over 70 students through a "flipped classroom", where students practice new problems each class and consult the TA for reinforced learning
- Designed and implemented weekly group problems and exam problems to test fundamental reaction engineering knowledge through unique and challenging problem sets
- Organized multiple in-person/remote office hours weekly and monthly review sessions

Hardware and Software Lab Manager | The Pennsylvania State University Jan. 2020 – Present

- Manage Janik lab hardware for efficient research through collaboration with PSU IT department
- Supervise supercomputing allocation usage across clusters for over 12 lab students
- Design and update the Dr. Janik lab website with research updates

Undergraduate Teaching Assistant | Texas A&M University Aug. 2018 – Dec. 2018
Course: Elementary Chemical Engineering

- Managed recitation of over 200 students to teach fundamental mass balance and chemical engineering principles through group practice problems

Private Lesson Teacher | Frisco Independent School District July 2015 – Aug. 2016

- Taught music and leadership principles to individuals or groups of over 50 students
- Managed lesson plans during leadership camp encompassing over 200 students

EXTRACURRICULAR AND LEADERSHIP ACTIVITIES

Penn State Chemical Engineering Graduate Student Association Aug. 2019 – Present

Mentor, Member

- Train and co-supervise undergraduate and Ph.D. students in using DFT methods for electrocatalysis, where both undergrads respectively were awarded NSF Fellowship and full-time job offer at Chemours
- Lead weekly electrochemistry meetings to facilitate collaboration for Ph.D. students across departments
- Mentoring first year Ph.D. students through courses and Ph.D. work, where one was awarded the best qualifying oral exam in 2022 under my supervision

American Institute of Chemical Engineers Mar. 2017 – Mar. 2019

Vice President, Publicity

- Awarded best national AIChE student chapter during VP tenure
- Coordinated record high meetings/workshop attendances for prospective and current ChE students
- Chaired session at the SPARK Conference to teach middle and high school students about engineering

Omega Chi Epsilon Chemical Engineering Honor Society Dec. 2017 – May 2019

Mentor, Member

- Invited to the organization to mentor engineering students due to outstanding academic achievement
- Assisted in annual Big Event to perform volunteer work for the community in Bryan, TX

HONORS AND AWARDS

- ACS ChemCatBio Award – March 2024
- NAM28 Kokes Award – Feb. 2023
- Dean's List – Jan. 2016 to May 2019

TECHNICAL SKILLS

Density Functional Theory, Vienna Ab Initio Simulation Package (VASP), Gaussian16, Python, Atomic Simulation Environment (ASE), Classical Molecular Dynamics, GROMACS, Amsterdam Modeling Suite , Materials Studio, VESTA, MATLAB, Apache Spark, TensorFlow, Wolfram Mathematica, and Microsoft Office Suite